Toward Robust Machine Learning based Network Intrusion Detection Systems

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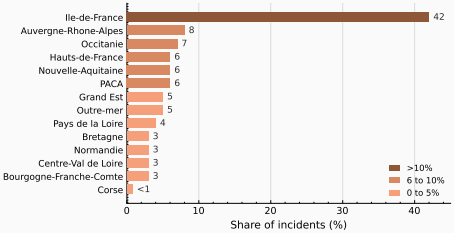
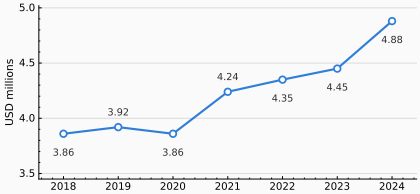
Introduction

Why this topic matters

- Modern societies increasingly rely on computer networks $\longrightarrow$ Higher exposure to cyber incidents.
- ANSSI 2024: **4,386** events handled (**+15%** vs. 2023); **1,361** confirmed incidents ($\approx$ **4/day**)
- Average data breach cost in 2024: **$4.88M**.
- In the French health sector, **35%** of reported incidents affected patient care.



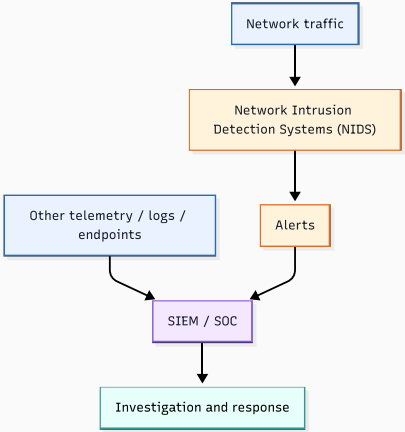
Source: **ANSSI, Panorama de la cybermenace 2024**



Source: **IBM Security, Cost of a Data Breach Report 2024**

Where NIDS fits in cyber defense

- Continuous monitoring to manage cyber risk
- SOCs and SIEM platforms centralize this monitoring
- Network traffic gives visibility into system communications
- NIDS turn traffic into alerts for investigation and response



⇒ **NIDS is one analytical layer inside the broader monitoring and response chain.**

From signatures to ML-based NIDS

Rule-based NIDS

- Tools: Snort, Suricata, Zeek
- Detect only known attack patterns or protocol violations, leaving unknown ones undetected.
- Encrypted traffic limits payload-based rules

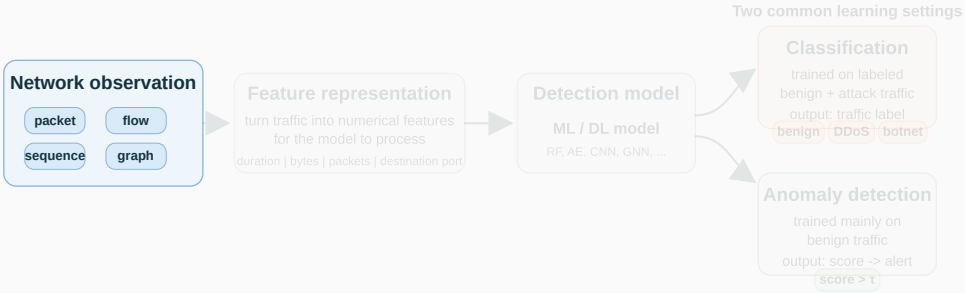
ML-based NIDS

- Example: Random Forest (RF) on flows
- Learn traffic patterns from historical data to generalize beyond exact rules.
- Can learn complex traffic patterns even under encryption

Operational reality: mostly hybrid[1]. E.g. SOCs use **Suricata/Zeek** for known alerts + **ML on flows** to flag suspicious HTTPS C&C behavior.

Ref: [1] SANS Detection & Response Survey (2024).

ML-based NIDS as a prediction task



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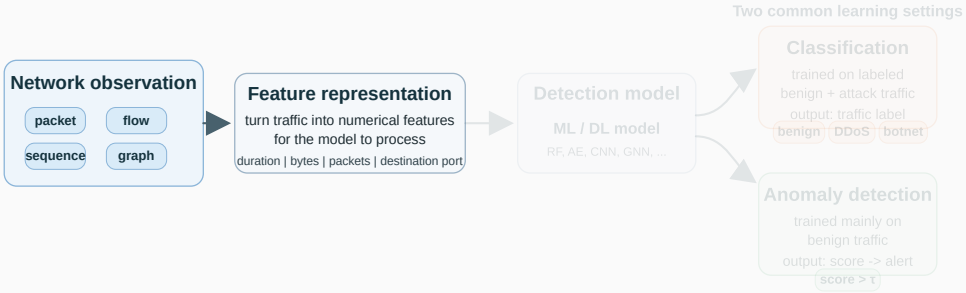

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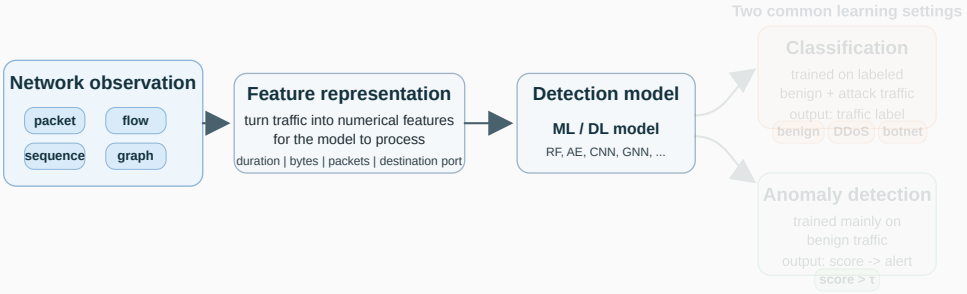

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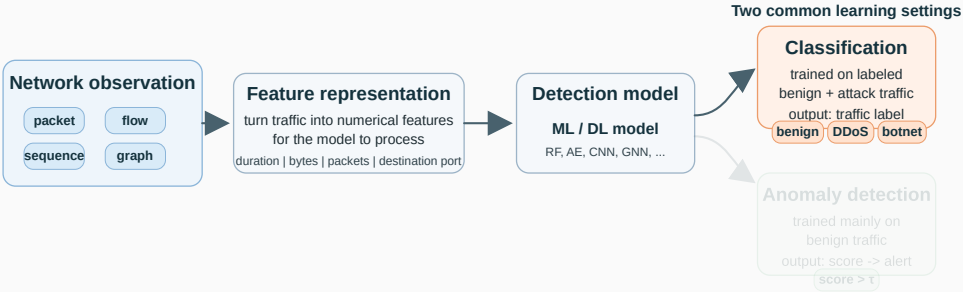

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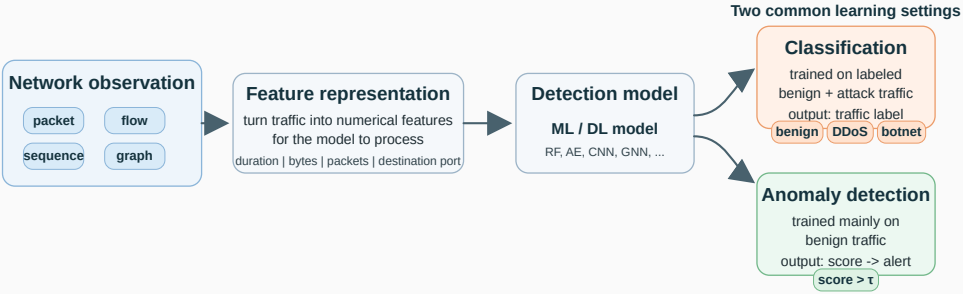

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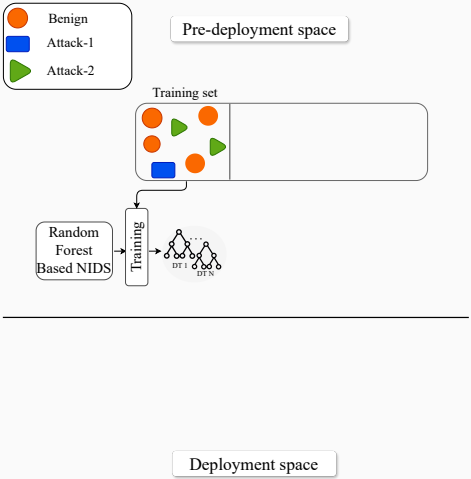

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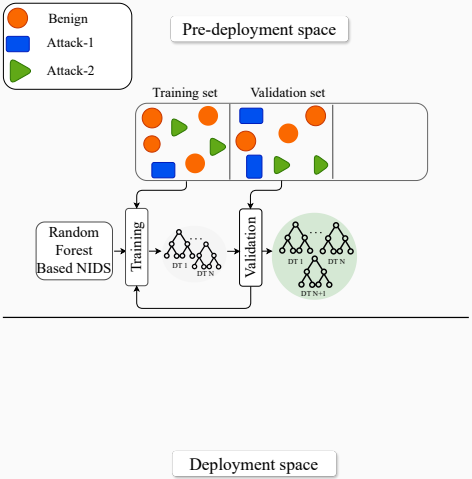

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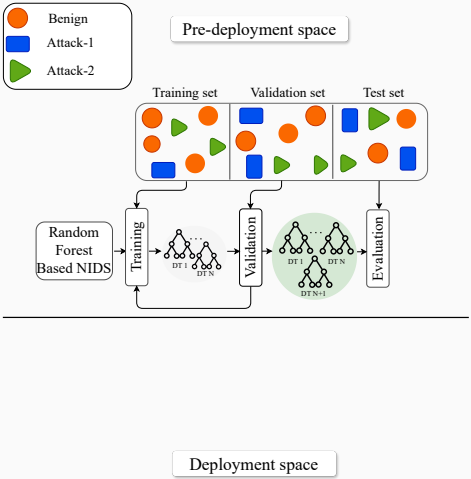

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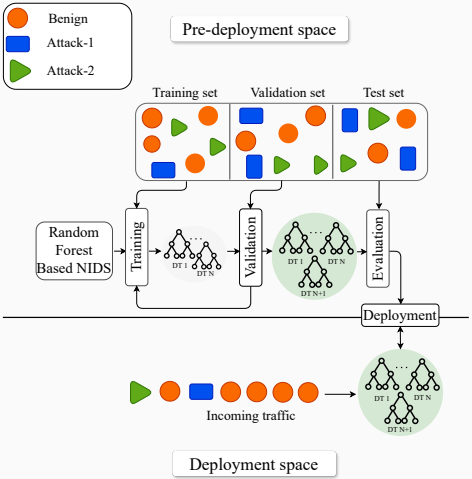
- The selected model is deployed to monitor live traffic.



How ML has been used to improve NIDS

- **A** Richer traffic representations ⟶ Flow-MAE[1], NetVigil[2]
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- **B** Better use of limited labels ⟶ Active-SVDD[3], INSOMNIA[4]
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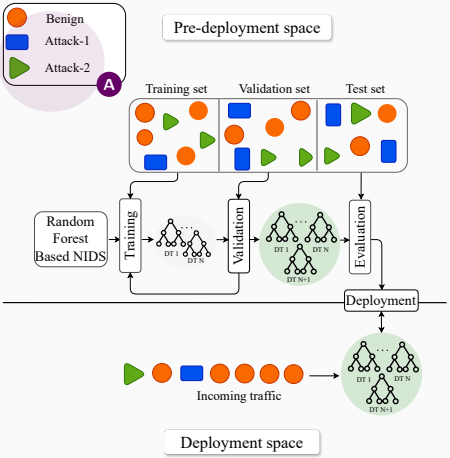

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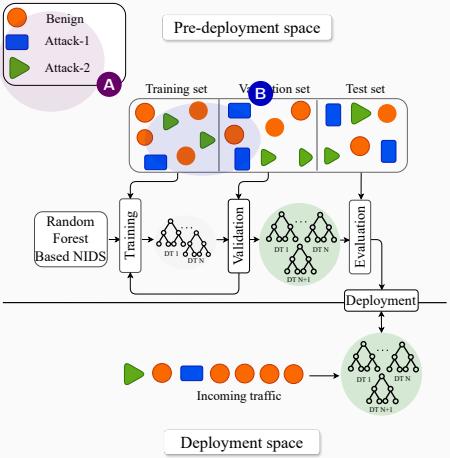

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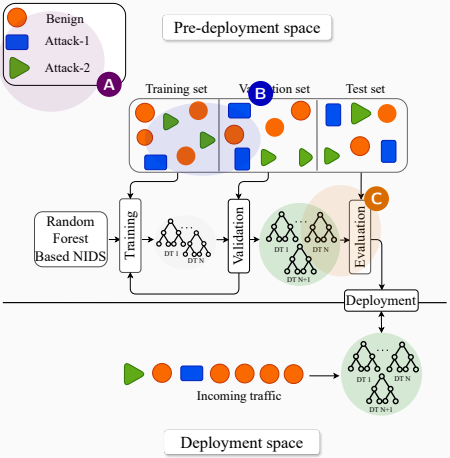

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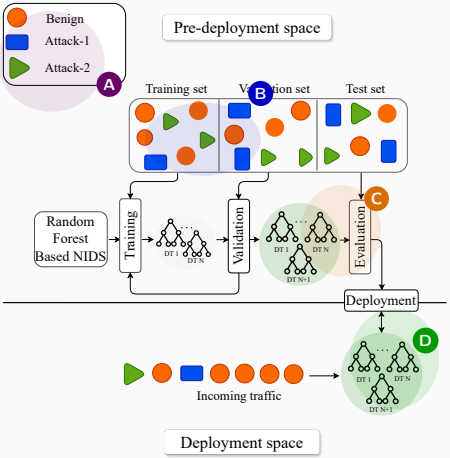

Refs: [7] AOC-IDS [Zhang et al., 2024, IEEE INFOCOM]; [8] ENIDrift [Wang, 2022, ACSAC].

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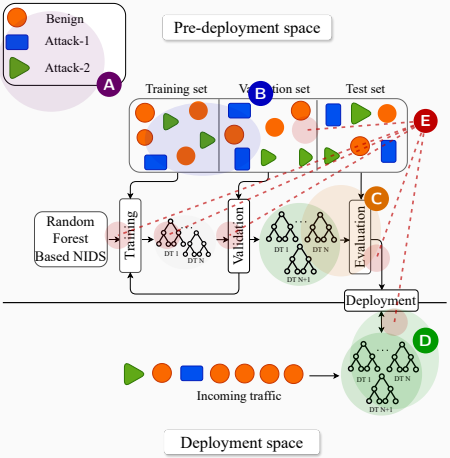

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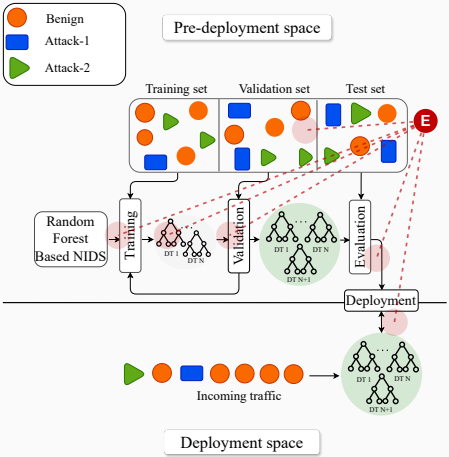

⇒ **The other axes optimize the detector inside a given setup, robustness questions the validity of that setup itself.**

Refs: [1] CADE [Yang et al., 2021, USENIX Security]; [2] OWAD [Han et al., 2023, NDSS].

Focus of this thesis: robustness in ML-based NIDS

Robustness in this thesis

- One fixed benchmark split is not enough.
- Robustness in operation: effectiveness under netwrok condition changes.
- Robustness in assessment: evaluation that truly stresses the detector.

$\Rightarrow$ **Robustness = adaptation + rigorous evaluation.**

Roadmap of the presentation

Part I Adaptation	Part II Evaluation
• Context change handling • Fast NIDS reconfiguration • Meta-learning-based auto-configuration	• Test-set difficulty assessment → relvealing NIDS weaknesses • Targeted augmentation of the test set when difficulty is low

Publications: Part I – *Auto-tuning of Hyper-parameters for Detecting Network Intrusions via Meta-learning* (NOMS 2023, AnNet);
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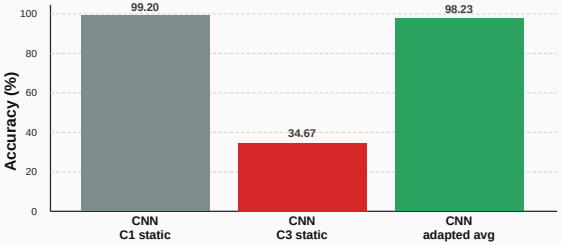
Why ML-based NIDS still need explicit adaptation

- Historical traffic for training; static detector at deployment.
- Assumes training and deployment traffic are statistically identical.
- Benign traffic evolves, attackers adapt → the network context changes.

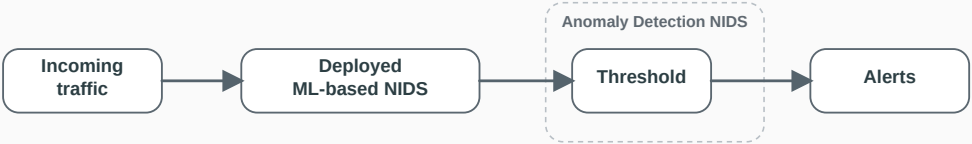


Source: Qiao et al., IEEE TII 2022, Table V/VI and Section IV-C

⇒ **Detection performance can degrade** ⇒ **explicit adaptation mechanisms are needed**

Adaptation = post-deployment update of (data | model | config | threshold) to maintain performance as the network context changes.

State of the art: main adaptation families



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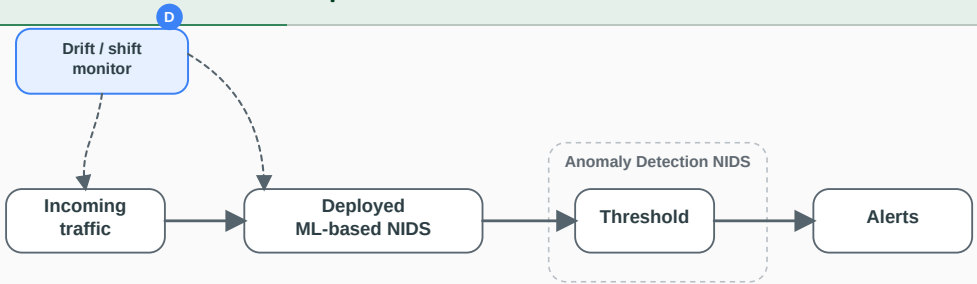

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monitor change in traffic or detector scores
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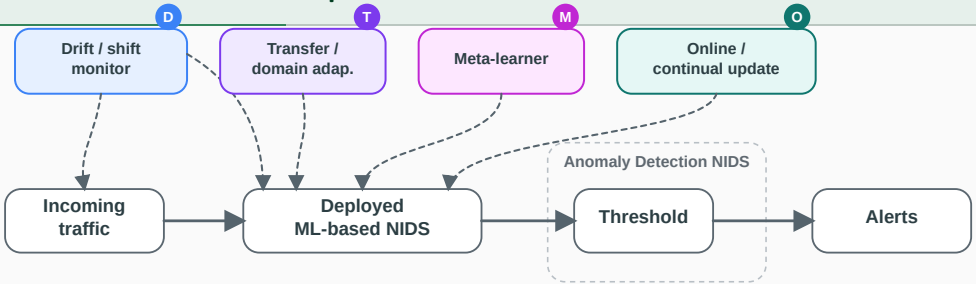

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ADA, NAEF, DI-NIDS

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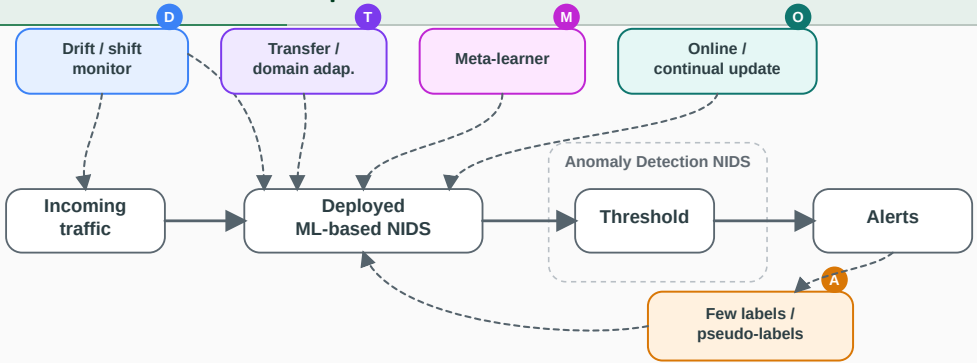

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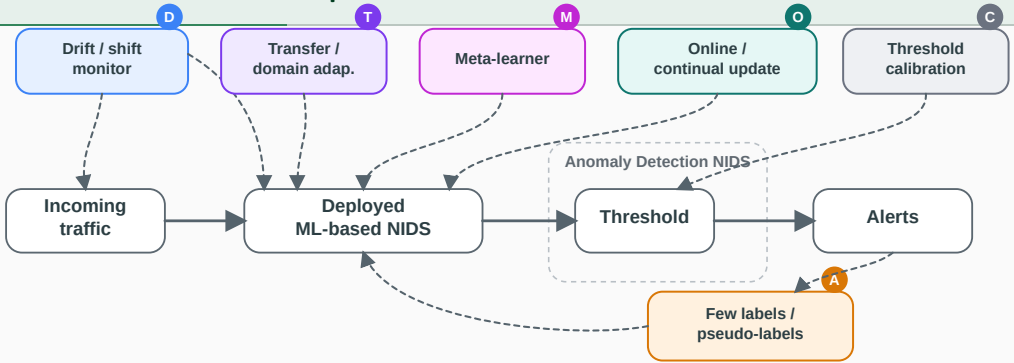

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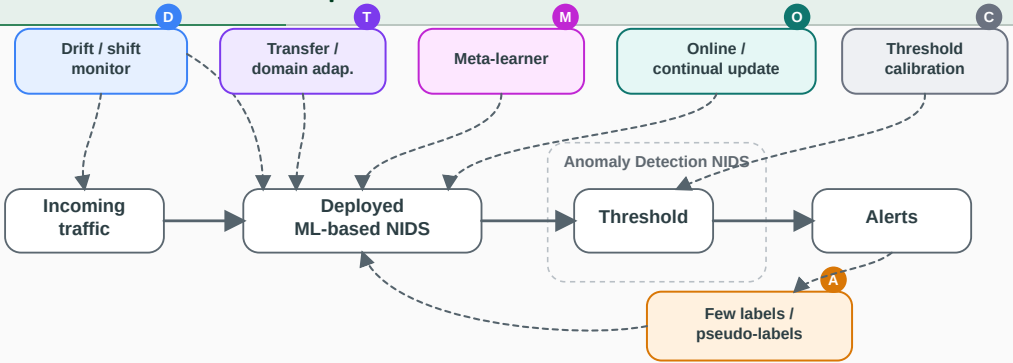

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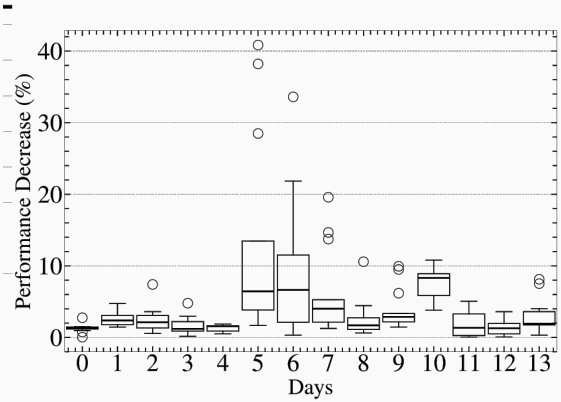
$\Rightarrow$ **Most methods adapt model, or threshold; configuration remains untargeted.**

Is hyperparameter configuration important?

- 14 daily IDS2017/IDS2018 datasets; each day is a separate context.
- Can a day-t optimal config be reused on day t'?
- Each boxplot $\rightarrow$ performance drop from transferring another day's best config.
- Limited loss on some days; 20–40% degradation on others.



Source: Anser et al., IEEE/IFIP NOMS 2026, Figure 5

$\Longrightarrow$ **Configuration transfer helps in some contexts, but is not reliable in general.**

How is configuration handled today?

Methods

- Grid search, random search[1], Bayesian optimization (BO)[2].
- Repeated train-and-validate over candidate configurations.

Main drawback

- Offline and expensive.
- Restarted from scratch after each context change.

Real NIDS example[3]

- Bayesian optimization + binary Random Forest NIDS.

300	10 s	$\approx$ 50 min
iterations	per iteration	per daily context

$\Longrightarrow$ **Too slow for repeated post-deployment reconfiguration.**

Research question (RQ1)

Research question

How can we find a configuration of a ML-based NIDS in a timely manner while maintaining an acceptable level of detection performance?

From our analysis of hyperparameter impact

- A NIDS configuration method has to be applied at each network context change.
- A new network context is defined by a new attack in the network or by a topology change.

Source: Anser et al., *An Empirical Study on Configuration Robustness of Unsupervised IDS*, IEEE/IFIP NOMS 2026.

$\Rightarrow$ **Auto-configuration via meta-learning**

Auto-configuration via Meta-learning

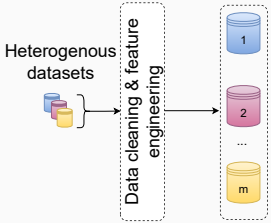
A. Meta-modeling

B. On the fly NIDS configuration

Trained NIDS

Auto-configuration via Meta-learning

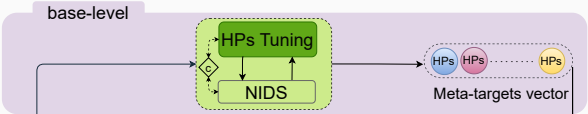
A. Meta-modeling

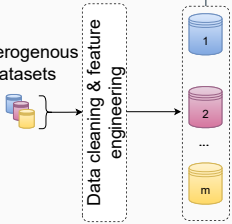


B. On the fly NIDS configuration

Auto-configuration via Meta-learning

For AD NIDS, tuning covers both HPs and threshold
HPs here = model HPs + threshold





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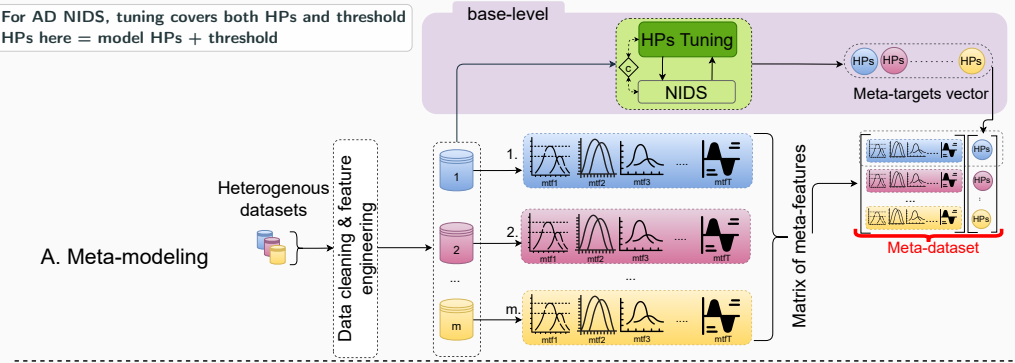

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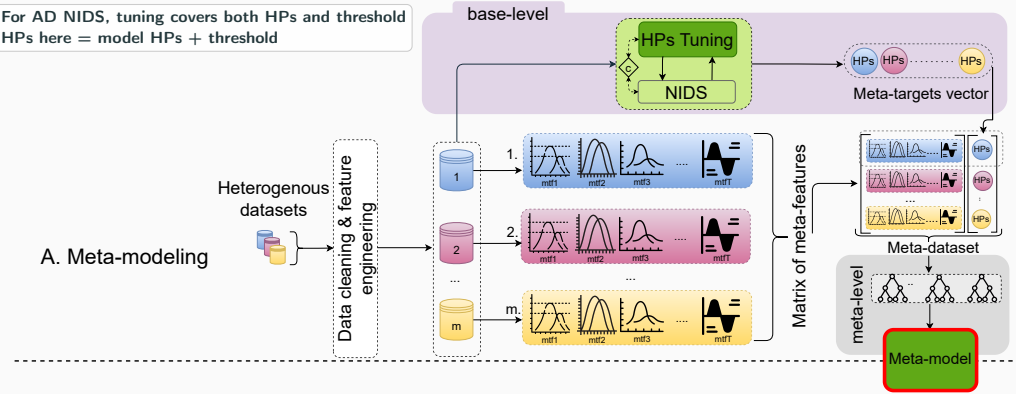

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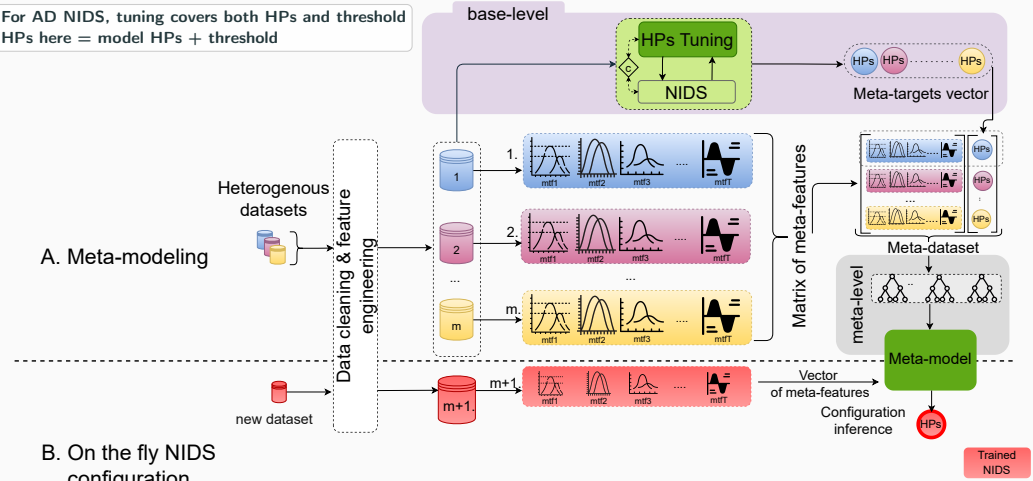

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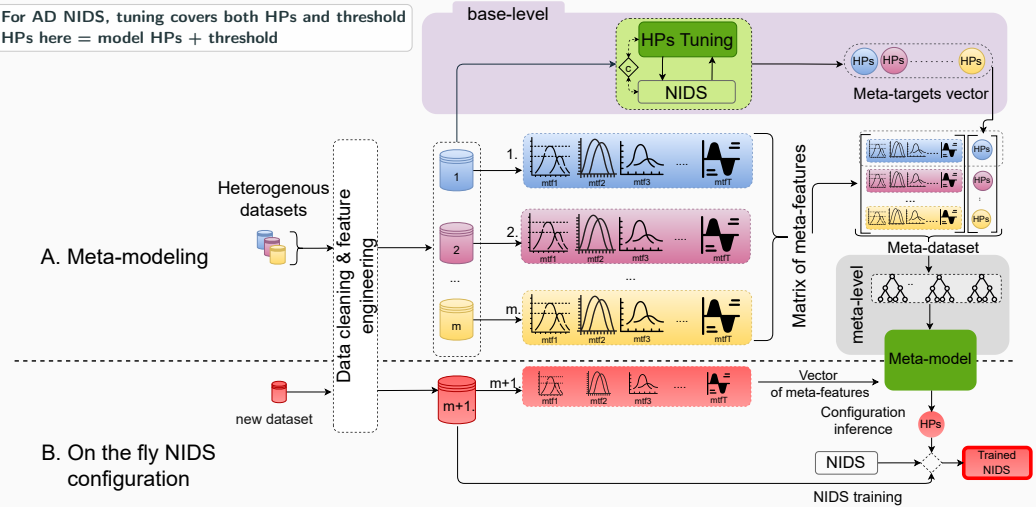


Methodology applied in two settings

Supervised setting

- Binary Random Forest on labeled benign/attack traffic.
- Contexts: **17 daily** IDS2017/IDS2018 datasets.
- Search space: **2 RF HPs** (‘max_features‘, ‘min_samples_leaf‘).
- BO baseline: **300 iterations/day**.

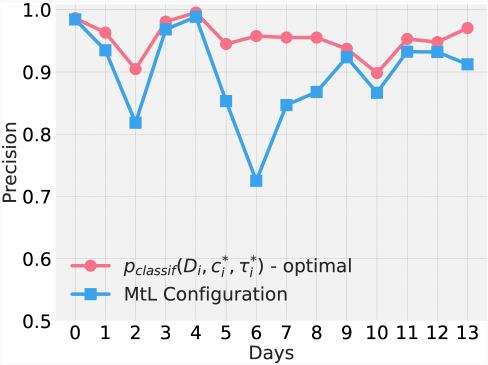
Unsupervised setting

- Isolation Forest-based AD IDS trained on attack-free data.
- Contexts: **14 daily** IDS2017/IDS2018 datasets.
- Search space: **7 iForest HPs + threshold**.
- BO baseline: offline daily optimum with labels.

$\Rightarrow$ **Same meta-learning pipeline, but different detectors, search spaces, and targets.**

Results: precision

- High overall precision in attack-free AD.
- Higher variability on days **2** and **5–8**.
- Harder contexts → weaker transfer.
- Supervised setting: meta-learning close to the (sub-)optimal configuration.

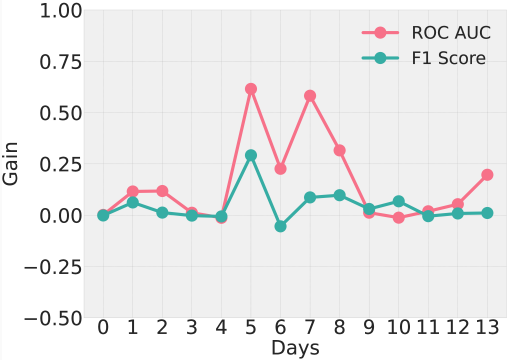


Unsupervised setting

⇒ **High precision overall; attack-free AD reveals the harder transfer days.**

Results: gain

- BO stopped at the same time budget as meta-learning.
- Gain = relative improvement: (meta-learning − baseline)/baseline.
- Strongest gains on harder days, especially **5–8**.
- Limited negative gains at equal cost.
- Supervised setting: large gains over the default.

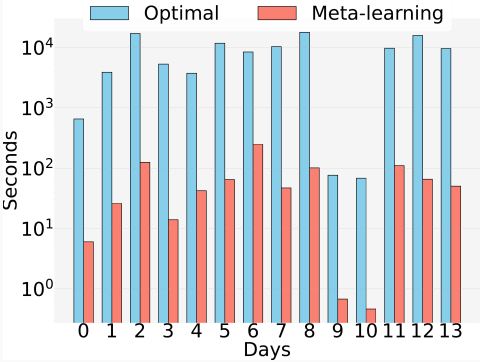


Unsupervised setting

⇒ **Cheaper than BO, yet competitive or better at equal budget.**

Results: configuration time

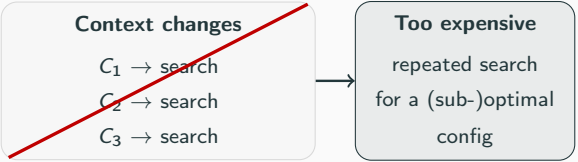
- Attack-free AD: seconds-to-minutes reconfiguration.
- BO runtime can exceed **100,000 s**.
- About **160x** faster overall.
- Supervised setting: about **9x** faster than finding a (sub-)optimal configuration.



Unsupervised setting

⇒ **Time advantage becomes decisive for online reconfiguration.**

Part I conclusion



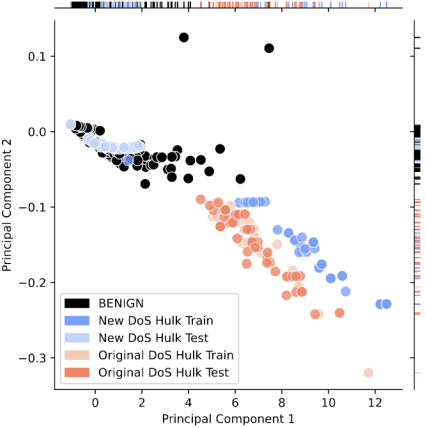
- Meta-learning preserves strong detection performance.
- The operational benefit is time: visible in supervised NIDS and decisive in attack-free AD.

⇒ **Detailed analysis: Anser et al., *An Empirical Study on Configuration Robustness of Unsupervised IDS*, IEEE/IFIP NOMS 2026.**

Part II: Evaluation

Benchmark NIDS datasets — A Closer Look

- A single source dataset partitioned into $\mathcal{D}_{\text{train}}$ and $\mathcal{D}_{\text{test}}$.
- Reported evaluation accuracies $>$ **90%** in published works on public benchmarks (IDS2017, UNSW-NB15...) $\rightarrow$ solved problem or benchmark-induced illusions?
- A near-complete overlap of DoS-Hulk flows in IDS2017's $\mathcal{D}_{\text{train}}$ and $\mathcal{D}_{\text{test}}$[1].

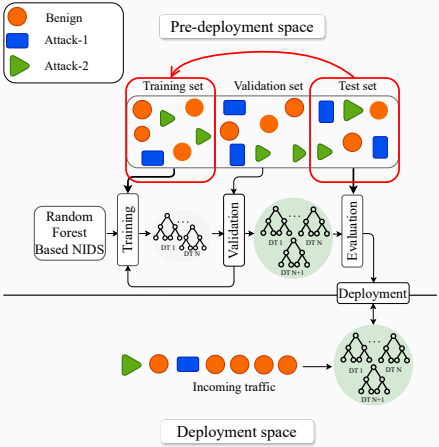


Overlap between $\mathcal{D}_{\text{test}}$ and $\mathcal{D}_{\text{train}}$ does not prevent overfitting $\rightarrow$ it can hide it.

[1] Flood et al., Bad Design Smells in Benchmark NIDS Datasets, IEEE EuroS&P 2024.

Focusing on $\mathcal{D}_{\text{test}}$ and its relation to $\mathcal{D}_{\text{train}}$

- A robust evaluation should expose challenging cases, including corner cases.
- Pre-deployment evaluation mainly relies on the test set $\mathcal{D}_{\text{test}}$.
- The key question is whether $\mathcal{D}_{\text{test}}$ really stresses the ML-based NIDS.
- The detector is constructed from $\mathcal{D}_{\text{train}} \rightarrow \mathcal{D}_{\text{test}}$ must be assessed relative to $\mathcal{D}_{\text{train}}$.



$\Rightarrow$ **This part focuses on $\mathcal{D}_{\text{test}}$ and its relation to $\mathcal{D}_{\text{train}}$.**

RQ2 — Targeted Augmentation

Given a **too-easy** initial $\mathcal{D}_{\text{test}}$, can we apply systematic augmentation to generate new, **realistic**, and **challenging** traffic that is specifically designed to expose the hidden weaknesses of the NIDS?

Research Questions

RQ1 — Test-set Assessment

How can we create a numerical score or a set of metrics to objectively measure the difficulty of a $\mathcal{D}_{\text{test}}$ for a NIDS?

RQ2 — Targeted Augmentation

Given a **too-easy** initial $\mathcal{D}_{\text{test}}$, can we apply systematic augmentation to generate new, **realistic**, and **challenging** traffic that is specifically designed to expose the hidden weaknesses of the NIDS?

Related Work

Method category	RQ1	RQ2	Key limitations
Neuron coverage			

Neuron coverage: Fraction of neurons whose activation exceeds a preset threshold at least once across the test set; higher → explore more of the model's internal logic.

Related Work

Method category	RQ1	RQ2	Key limitations
Neuron coverage			
Surprise coverage			

Surprise coverage: Measures how broadly the test set differs from what the model saw in training; higher → the test set includes highly surprising inputs.

Related Work

Method category	RQ1	RQ2	Key limitations
Neuron coverage			
Surprise coverage			
Mutation testing			

Mutation testing: Mutates the model and/or its training setup (e.g., weights, architecture, or hyperparameters), then checks if the test set can distinguish the mutant from the original model.

Related Work

Method category	RQ1	RQ2	Key limitations
Neuron coverage	✓	✗	Model access (🔓) + NN-only ( ) + Ignores train–test alignment (📏)
Surprise coverage	✓	✗	🔓 + + Proximity-only focus (🎯)
Mutation testing	✓	✓	🔓 + 📏 + Model-change sensitive (🛠) Retrain per dataset (🔄) + Synthetic data (🧪)
Manual (Flood et al., 2024)	✗	✓	Unguided augmentation (👆)

G1 Model-related gaps **G2** Shortcomings in assessing $\mathcal{D}_{\text{test}}$ → RQ1 **G3** Weaknesses in $\mathcal{D}_{\text{test}}$ augmentation workflows → RQ2

TATA (Test-set Assessment & Targeted Augmentation)

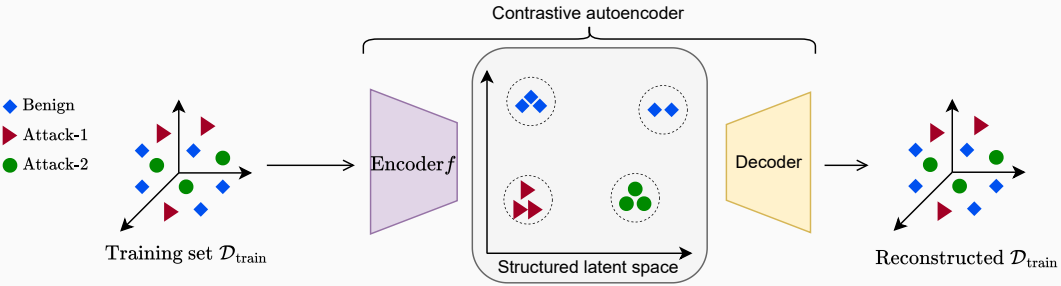
Goal: Close the gaps identified in related work and answer the research questions.

1. **Preliminary Phase** G1 : + +
   Approximate $\mathcal{D}_{\text{train}}$'s decision boundary while being model agnostic

2. **Phase 1 – Assesses $\mathcal{D}_{\text{test}}$'s quality (RQ1)** G2 : +
   Define three metrics (**Proximity**, **Diversity**, and **Scarcity**) to assess the **alignment** of $\mathcal{D}_{\text{test}}$ with respect to $\mathcal{D}_{\text{train}}$'s distribution.

3. **Phase 2 – $\mathcal{D}_{\text{test}}$ Augmentation (RQ2)** G3 : + +
   Reinforcement learning (RL) guides a testbed to generate **realistic traffic** that fills uncovered metric gaps, producing **a more challenging** $\mathcal{D}_{\text{test}}$.

Preliminary Phase: Approximating $\mathcal{D}_{\text{train}}$'s decision boundary



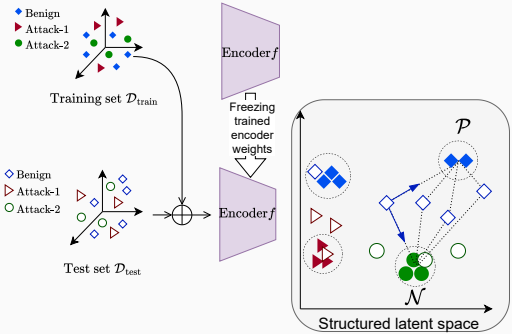
- Decision boundaries $\approx$ cluster borders in the latent space; these are captured in the encoder's weights.

Phase 1: Test-set Assessment

- Encode $\mathcal{D}_{\text{train}}$ & $\mathcal{D}_{\text{test}} \rightarrow \mathcal{Z}_{\text{test}}$ is located relative to $\mathcal{Z}_{\text{train}} \rightarrow$ we now work entirely in latent space $(\mathcal{Z}_{\text{train}}, \mathcal{Z}_{\text{test}})$.

- Proximity, diversity, and scarcity to characterize where $\mathcal{Z}_{\text{test}}$ lies in latent space relative to $\mathcal{Z}_{\text{train}}$ clusters.

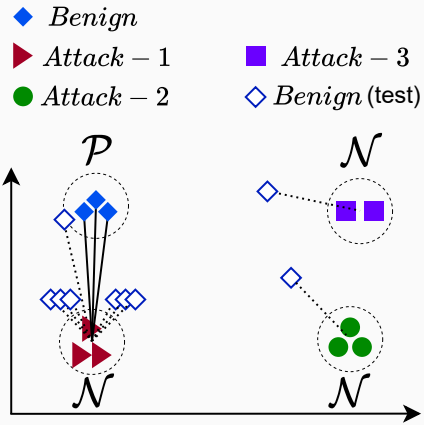
- For each test embedding z, we identify:
  - $P(z)$: the nearest **same-label cluster** (*positive cluster*)
  - $N(z)$: the nearest **different-label cluster** (*negative cluster*)



Proximity $P \in [0, 1]$, higher = nearer to different-label boundaries

- Quantifies $\mathcal{Z}_{\text{test}}$ closeness to $N(z)$; high proximity $\Rightarrow$ closer to a different label decision boundary $\Rightarrow$ harder cases.

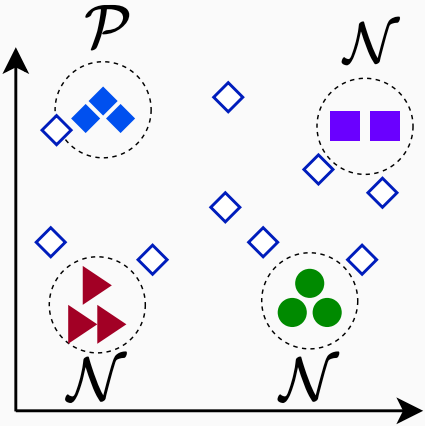
- P = **max** one-sided Kolmogorov–Smirnov statistic (over clusters): compares $\mathcal{Z}_{\text{test}}$ vs. $\mathcal{Z}_{\text{train}}$ distances to $N(z)$



Diversity $D \in [0, 1]$, higher = broader, less redundant coverage

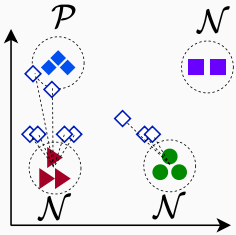
- Assesses the variety of $\mathcal{Z}_{\text{test}}$; higher diversity $\Rightarrow$ minimal redundancy $\Rightarrow$ broader coverage of the latent space.

- D = **Average** of normalized **Vendi Scores** (over clusters)

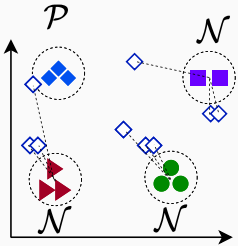


Scarcity $S \in [0, 1]$ higher = boundaries challenged more evenly

- Evaluates how uniformly $\mathcal{Z}_{\text{test}}$ are distributed among all potential negative clusters (i.e., decision boundaries).
- High scarcity $\Rightarrow$ a uniform spread $\Rightarrow$ all boundaries are exercised $\Rightarrow$ a more robust test set.
- S = **Average** of the **(1 - Gini coefficient)** (over clusters)

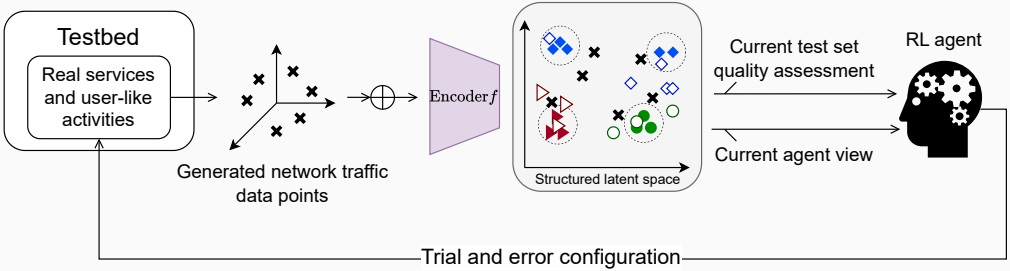


Bad scarcity



Optimal scarcity

Phase 2: Targeted Augmentation



- Start from Preliminary & Phase 1 outputs
- Configurable live testbed for realistic traffic generation.
- Legitimate multi-protocol traffic (HTTP/HTTPS, SSH, SMTP, DNS, . . .) and controlled attack traffic.

- **Training workflow**:

$$\boxed{\text{observe latent-space}} \rightarrow \text{select a configuration} \rightarrow \text{traffic generation} \xrightarrow{\text{embed}} \text{update } \mathcal{Z}_{\text{test}}$$

$$\rightarrow \text{compute proximity, diversity, and scarcity} \rightarrow \text{update policy } \pi \circlearrowleft \Rightarrow \pi^*$$

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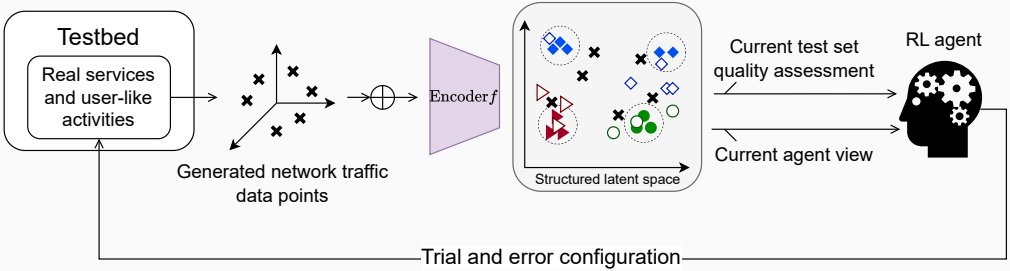

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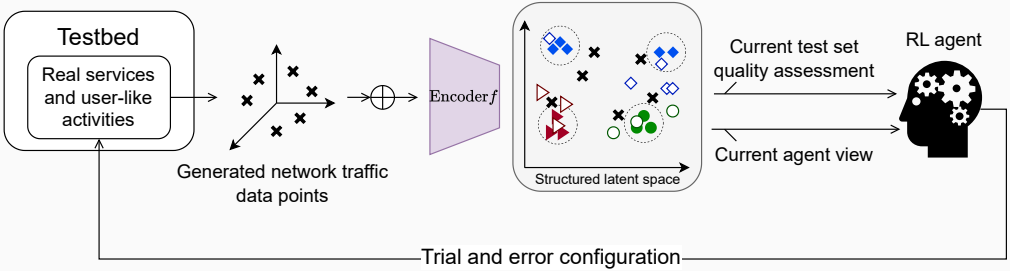

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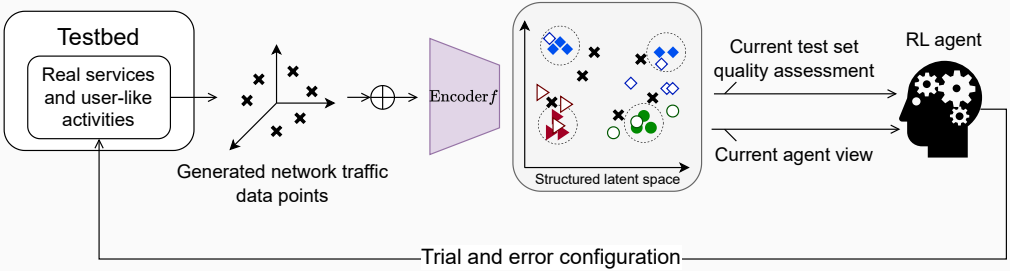

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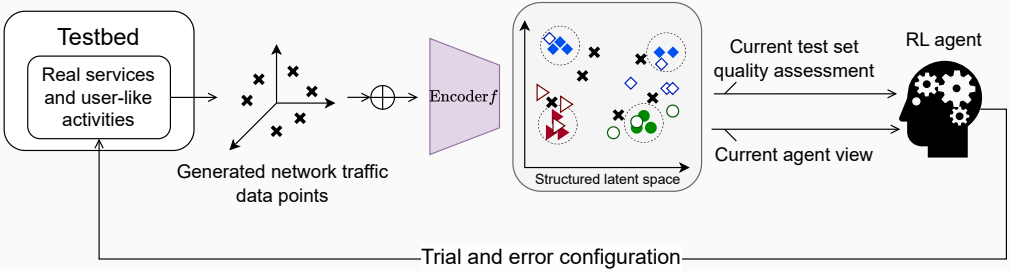

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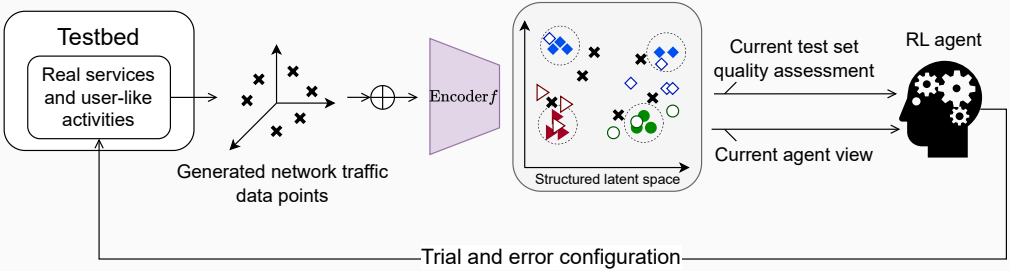

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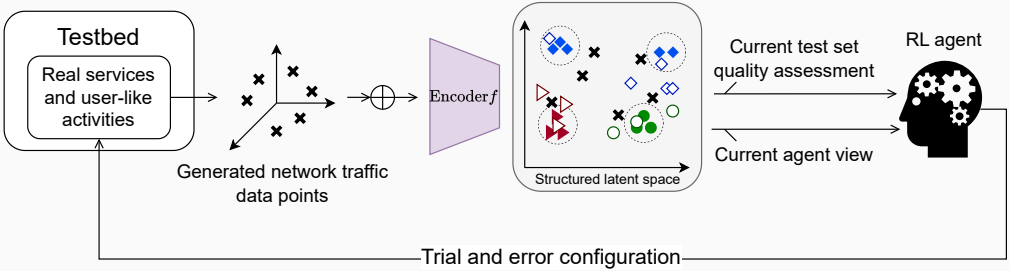

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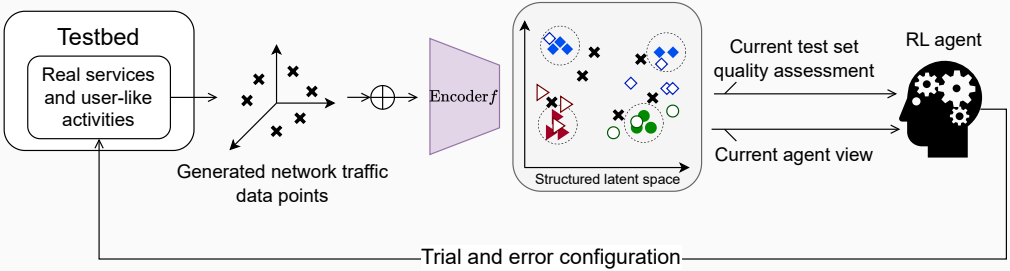

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Experimental Setup

Datasets (60 / 20 / 20 split)

(Using the revised and corrected versions)

- **CIC-IDS 2017**: 0.93 M benign / 0.25 M attack flows
- **CSE-CIC-IDS 2018**: 59.8 M benign / 4.1 M attack flows

NIDS Models

- RF, SVM, 3-layer DNN (128-128-128 ReLU + Softmax)

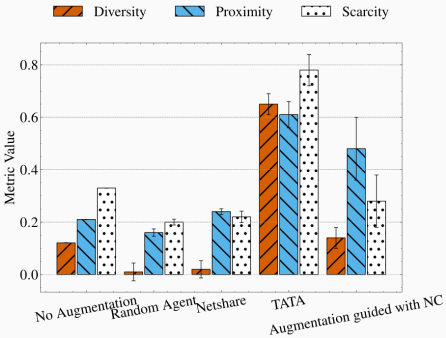
Agent Training

- Augmentation targets Benign SSH flows only

- Action space (client-side via `tc`): **Loss** $\in [5, 10]\%$ | **Jitter** $\in [4, 10]\,\mathrm{ms}$ | **Delay** $\in [10, 40]\,\mathrm{ms}$ | **Duplication** $\in [0.1, 5]\%$ | **Corruption** $\in [0.1, 10]\%$ | **Reordering** $\in [0.1, 50]\%$ | **Correlation** $\in [50, 100]\%$.

Test-set Augmentation — Metrics

- **Protocol:** 20 eval episodes on IDS2017; no reward/updates; add Benign SSH flows; report mean±sd.
- **Baselines:** Random, NetShare (GAN), RL + Neuron Coverage (NC).



Test-set quality metrics (higher is more challenging/covered).

Test-set Augmentation — Metrics

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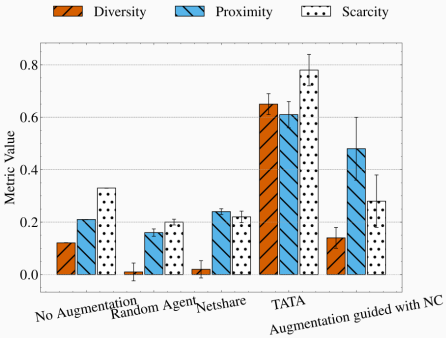
  - **TATA:** $D + \sim 437\%$; $P + \sim 190\%$; $S + \sim 136\%$
  - **NC:** $D \approx 0$; $P + \sim 129\%$; $S \approx 0$
  - **Random/NetShare:** $D \downarrow$ to ≈ 0; $P \approx 0$; $S \downarrow$



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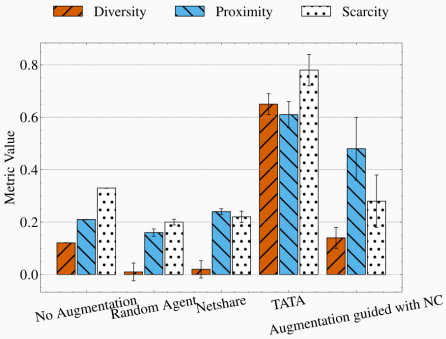
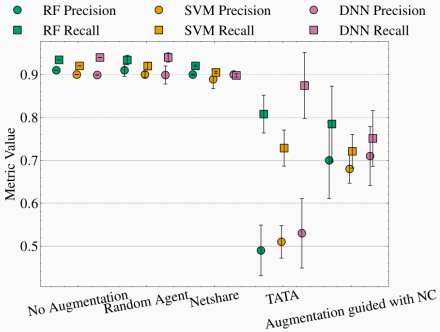


Test-set quality metrics (higher is more challenging/covered).

What is the impact on NIDS performance?

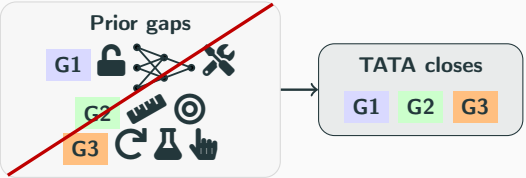
Test-set Augmentation — Impact on NIDS

- **NIDS impact (RF example):**
  - **TATA:** Precision $-\sim 46\%$; Recall $-\sim 13\%$
  - **NC:** Precision $-\sim 23\%$; Recall $-\sim 16\%$
- **Note:** higher scarcity spreads errors across boundaries $\Rightarrow$ FP $\uparrow \Rightarrow$ precision $\downarrow$.
- $\Rightarrow$ **proximity is the key driver;** scarcity and diversity are complementary (spread errors, avoid redundancy)

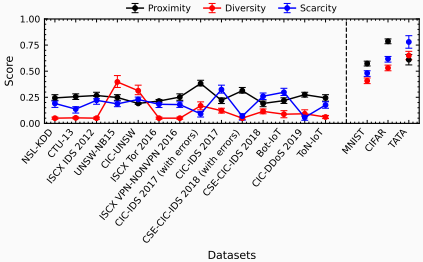


Impact on detection metrics (example: RF).

Part II conclusion



- Across public NIDS benchmarks, metrics remain mostly below **0.5** ⇒ limited test-set difficulty.



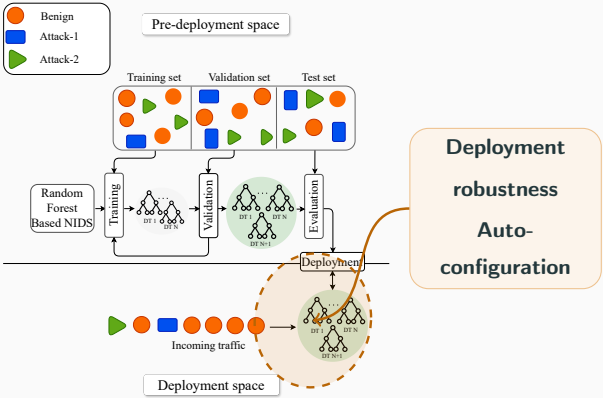
Public NIDS benchmarks ordered on the x-axis by release year (2009–2020)

Conclusion and perspectives

General conclusion

Auto-configuration perspectives

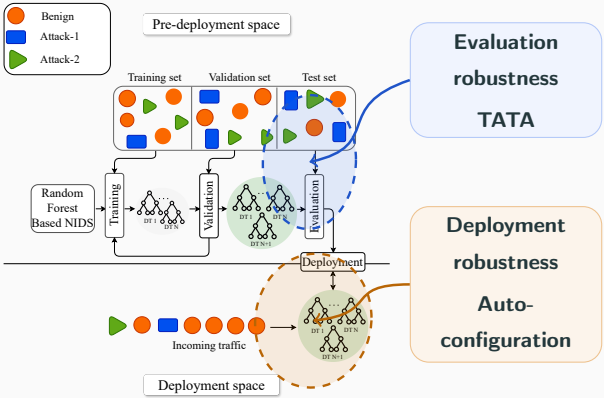
- Closed-loop reconfiguration under context change.
- Cold-start mitigation across organizations. → privacy-preserving and federated configuration learning.



General conclusion

TATA perspectives

- Reuse hard cases for detector hardening.
- Beyond single-protocol, flow-based settings.
- Larger and richer stress-test testbeds.



$\Rightarrow$ **Robustness in this thesis = rigorous evaluation + adaptive configuration.**

Domain Perspectives

General takeaway

- Robust ML-based NIDS require both stronger pre-deployment evaluation and adaptive post-deployment configuration.

Perspectives

- Move from one-shot benchmarking to continuous evaluation and adaptation loops.
- Use GenAI for scenario-to-traffic generation and analyst assistance, while accounting for more adaptive attackers.

Questions?